MSG No. 194.265 $P6_3/m'mc$ [Type III, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-3'm.`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,3,5,7,8,9,13,14,15,19,21,23]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[2,4,6,10,11,12,16,17,18,20,22,24]`

Table 2: Wyckoff site: `2b`, site symmetry: `-6'm2'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,3,5,7,8,9,16,17,18,20,22,24]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[2,4,6,10,11,12,13,14,15,19,21,23]`

Table 3: Wyckoff site: `2c`, site symmetry: `-6'm2'`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[1,3,5,7,8,9,16,17,18,20,22,24]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[2,4,6,10,11,12,13,14,15,19,21,23]`

Table 4: Wyckoff site: `2d`, site symmetry: `-6'm2'`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[1,3,5,7,8,9,16,17,18,20,22,24]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[2,4,6,10,11,12,13,14,15,19,21,23]`

Table 5: Wyckoff site: `4e`, site symmetry: `3m.`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,3,5,7,8,9]`
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[2,4,6,10,11,12]`
3	$[0,\ 0,\ -z]$	`[13,14,15,19,21,23]`
4	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[16,17,18,20,22,24]`

Table 6: Wyckoff site: `4f`, site symmetry: `3m.`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,3,5,7,8,9]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[2,4,6,10,11,12]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[13,14,15,19,21,23]`
4	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	`[16,17,18,20,22,24]`

Table 7: Wyckoff site: `6g`, site symmetry: `.2'/m.`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,7,13,19]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,10,16,20]`
3	$\left[0,\ \frac{1}{2},\ 0\right]$	`[3,8,14,21]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[4,11,17,22]`
5	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[5,9,15,23]`
6	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[6,12,18,24]`

Table 8: Wyckoff site: `6h`, site symmetry: `m'm2'`

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{1}{4}\right]$	`[1,7,17,22]`
2	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[2,10,15,23]`
3	$\left[-2x,\ -x,\ \frac{1}{4}\right]$	`[3,8,18,24]`
4	$\left[-x,\ -2x,\ \frac{3}{4}\right]$	`[4,11,13,19]`
5	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[5,9,16,20]`
6	$\left[2x,\ x,\ \frac{3}{4}\right]$	`[6,12,14,21]`

Table 9: Wyckoff site: `12i`, site symmetry: `.2'.`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,13]`
2	$\left[x,\ x,\ \frac{1}{2}\right]$	`[2,16]`
3	$[0,\ x,\ 0]$	`[3,14]`
4	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[4,17]`
5	$[-x,\ -x,\ 0]$	`[5,15]`
6	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[6,18]`
7	$[-x,\ 0,\ 0]$	`[7,19]`
8	$[0,\ -x,\ 0]$	`[8,21]`
9	$[x,\ x,\ 0]$	`[9,23]`
10	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[10,20]`
11	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[11,22]`
12	$\left[0,\ x,\ \frac{1}{2}\right]$	`[12,24]`

Table 10: Wyckoff site: `12j`, site symmetry: `m'..`

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{4}\right]$	`[1,22]`
2	$\left[x-y,\ x,\ \frac{3}{4}\right]$	`[2,23]`
3	$\left[-y,\ x-y,\ \frac{1}{4}\right]$	`[3,24]`
4	$\left[-x,\ -y,\ \frac{3}{4}\right]$	`[4,19]`
5	$\left[-x+y,\ -x,\ \frac{1}{4}\right]$	`[5,20]`
6	$\left[y,\ -x+y,\ \frac{3}{4}\right]$	`[6,21]`
7	$\left[-x+y,\ y,\ \frac{1}{4}\right]$	`[7,17]`

continued ...

Table 10

No.	position	mapping
8	$\left[-y,\ -x,\ \frac{1}{4}\right]$	[8,18]
9	$\left[x,\ x-y,\ \frac{1}{4}\right]$	[9,16]
10	$\left[-x,\ -x+y,\ \frac{3}{4}\right]$	[10,15]
11	$\left[x-y,\ -y,\ \frac{3}{4}\right]$	[11,13]
12	$\left[y,\ x,\ \frac{3}{4}\right]$	[12,14]

Table 11: Wyckoff site: 12k, site symmetry: .m.

No.	position	mapping
1	$[x,\ 2x,\ z]$	[1,7]
2	$\left[-x,\ x,\ z+\frac{1}{2}\right]$	[2,10]
3	$[-2x,\ -x,\ z]$	[3,8]
4	$\left[-x,\ -2x,\ z+\frac{1}{2}\right]$	[4,11]
5	$[x,\ -x,\ z]$	[5,9]
6	$\left[2x,\ x,\ z+\frac{1}{2}\right]$	[6,12]
7	$[-x,\ -2x,\ -z]$	[13,19]
8	$[2x,\ x,\ -z]$	[14,21]
9	$[-x,\ x,\ -z]$	[15,23]
10	$\left[x,\ -x,\ \frac{1}{2}-z\right]$	[16,20]
11	$\left[x,\ 2x,\ \frac{1}{2}-z\right]$	[17,22]
12	$\left[-2x,\ -x,\ \frac{1}{2}-z\right]$	[18,24]

Table 12: Wyckoff site: 24l, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x-y,\ x,\ z+\frac{1}{2}\right]$	[2]
3	$[-y,\ x-y,\ z]$	[3]
4	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[4]
5	$[-x+y,\ -x,\ z]$	[5]
6	$\left[y,\ -x+y,\ z+\frac{1}{2}\right]$	[6]
7	$[-x+y,\ y,\ z]$	[7]
8	$[-y,\ -x,\ z]$	[8]
9	$[x,\ x-y,\ z]$	[9]
10	$\left[-x,\ -x+y,\ z+\frac{1}{2}\right]$	[10]
11	$\left[x-y,\ -y,\ z+\frac{1}{2}\right]$	[11]
12	$\left[y,\ x,\ z+\frac{1}{2}\right]$	[12]
13	$[x-y,\ -y,\ -z]$	[13]
14	$[y,\ x,\ -z]$	[14]
15	$[-x,\ -x+y,\ -z]$	[15]
16	$\left[x,\ x-y,\ \frac{1}{2}-z\right]$	[16]
17	$\left[-x+y,\ y,\ \frac{1}{2}-z\right]$	[17]
18	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[18]
19	$[-x,\ -y,\ -z]$	[19]

continued ...

Table 12

No.	position	mapping
20	$\left[-x+y,\ -x,\ \frac{1}{2}-z\right]$	[20]
21	$[y,\ -x+y,\ -z]$	[21]
22	$\left[x,\ y,\ \frac{1}{2}-z\right]$	[22]
23	$[x-y,\ x,\ -z]$	[23]
24	$\left[-y,\ x-y,\ \frac{1}{2}-z\right]$	[24]